

## Ejercicios de practica

1. Serie  $S = 1/2 + 2/2^2 + 3/3^2 + \dots + n/n^2$

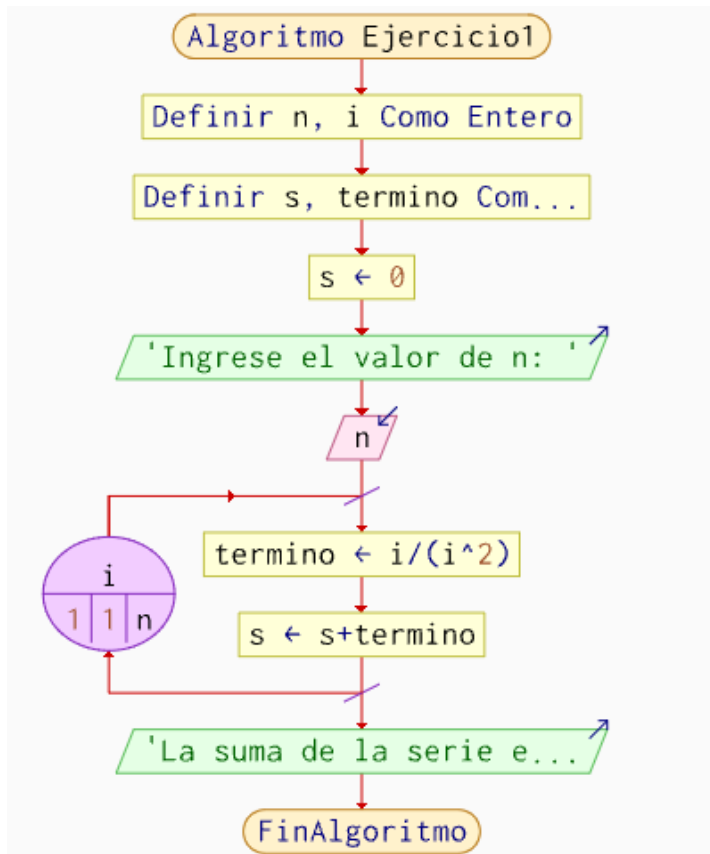
```
using System;|
0 referencias
class Ejercicio1
{
    0 referencias
    static void Main()
    {
        Console.WriteLine("Ingrese el valor de n: ");
        int n = int.Parse(Console.ReadLine());

        double s = 0;
        for (int i = 1; i <= n; i++)
        {
            double termino = (double) i / (i * i);
            s += termino;
            Console.WriteLine($"Termino {i}: {i}/{i}^2 = {termino}");
        }

        Console.WriteLine($"La suma de la serie es S = {s}");
        Console.ReadKey();
    }
}
```

```
Ingrese el valor de n: 10
Termino 1: 1/1^2 = 1
Termino 2: 2/2^2 = 0,5
Termino 3: 3/3^2 = 0,3333333333333333
Termino 4: 4/4^2 = 0,25
Termino 5: 5/5^2 = 0,2
Termino 6: 6/6^2 = 0,16666666666666666
Termino 7: 7/7^2 = 0,14285714285714285
Termino 8: 8/8^2 = 0,125
Termino 9: 9/9^2 = 0,11111111111111111
Termino 10: 10/10^2 = 0,1

La suma de la serie es S = 2,9289682539682538
```



2. Serie de factoriales: 1, 2, 6, 24, 120, 720, ...

```

using System;
0 referencias
class Ejercicio2
{
    0 referencias
    static void Main()
    {
        Console.WriteLine("Ingrese el valor de n: ");
        int n = int.Parse(Console.ReadLine());

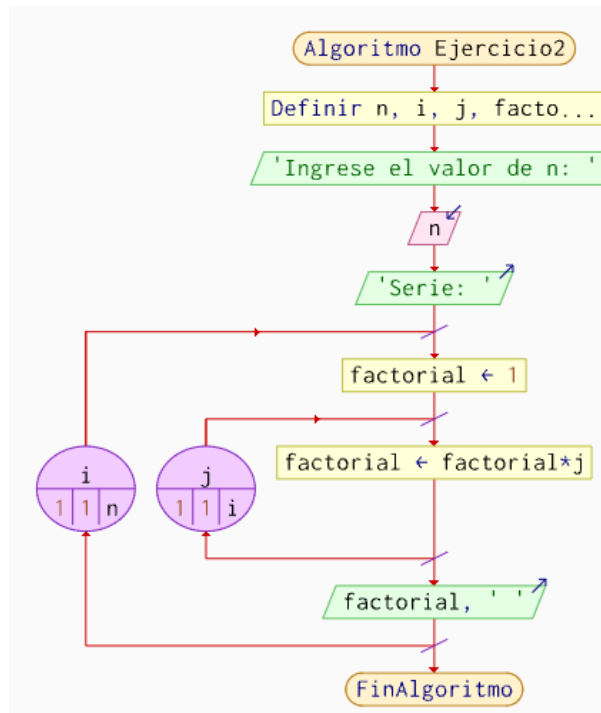
        Console.WriteLine("Serie: ");
        for (int i = 1; i <= n; i++)
        {
            long factorial = 1;
            for (int j = 1; j <= i; j++)
                factorial *= j;

            Console.WriteLine(factorial);
            if (i < n) Console.Write(", ");
        }
        Console.WriteLine();
        Console.ReadKey();
    }
}
  
```

Visualización de la salida del programa en una ventana de consola:

```

C:\Users\USUARIO\Desktop\...
Ingrese el valor de n: 10
Serie: 1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800
  
```



3. Serie 1, 2, 5, 10, 17, 26, 37, ...

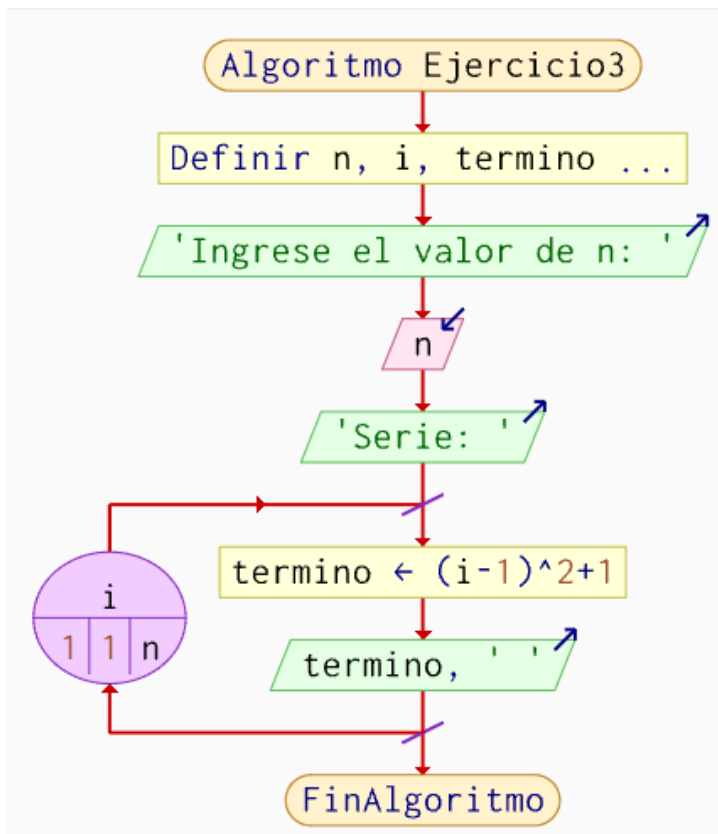
```

using System;
0 referencias
class Ejercicio3
{
    0 referencias
    static void Main()
    {
        Console.WriteLine("Ingrese el valor de n: ");
        int n = int.Parse(Console.ReadLine());

        Console.WriteLine("Serie: ");
        for (int i = 1; i <= n; i++)
        {
            int termino = (i - 1) * (i - 1) + 1;
            Console.Write(termino);
            if (i < n) Console.Write(", ");
        }
        Console.WriteLine();
        Console.ReadKey();
    }
}
  
```

C:\Users\USUARIO\Desktop\e x + - □ x

Ingrese el valor de n: 10  
 Serie: 1, 2, 5, 10, 17, 26, 37, 50, 65, 82



4. Serie 2, 3, 5, 8, 12, 17, 23, ..

```

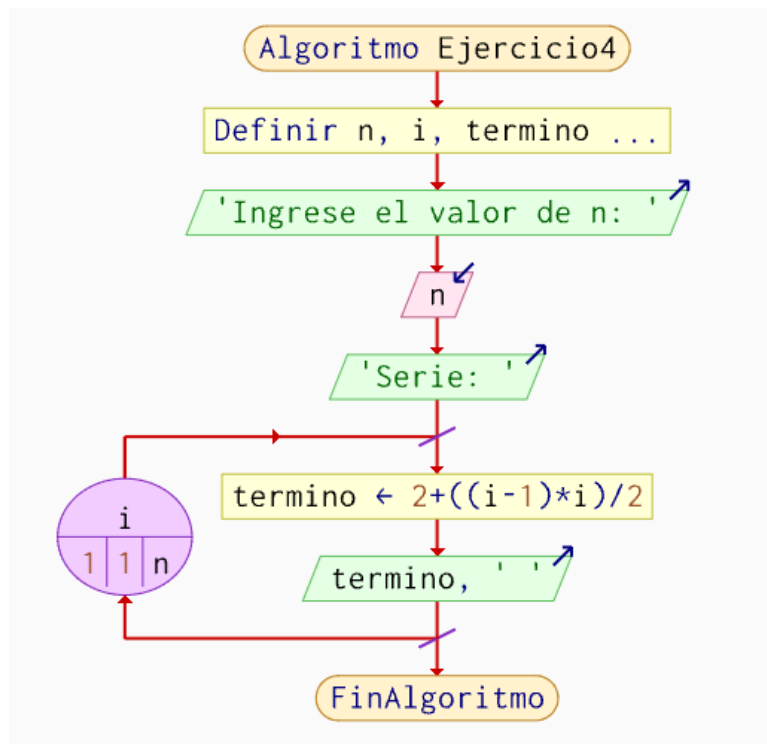
1  using System;
2  0 referencias
3  class Ejercicio4
4  {
5      0 referencias
6      static void Main()
7      {
8          Console.WriteLine("Ingrese el valor de n: ");
9          int n = int.Parse(Console.ReadLine());
10
11          Console.WriteLine("Serie: ");
12          for (int i = 1; i <= n; i++)
13          {
14              int termino = 2 + (i - 1) * i / 2;
15              Console.Write(termino);
16              if (i < n) Console.Write(", ");
17          }
18          Console.WriteLine();
19          Console.ReadKey();
20      }
21  }

```

Minimizar

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Ingrese el valor de n: 10  
Serie: 2, 3, 5, 8, 12, 17, 23, 30, 38, 47



5. Serie 1, 22, 333, 4444, 55555, ...

```

1  using System;
2  class Ejercicio5
3  {
4      static void Main()
5      {
6          Console.WriteLine("Ingrese el valor de n: ");
7          int n = int.Parse(Console.ReadLine());
8
9          Console.WriteLine("Serie: ");
10         for (int i = 1; i <= n; i++)
11         {
12             string termino = "";
13             for (int j = 1; j <= i; j++)
14                 termino += i.ToString();
15
16             Console.Write(termino);
17             if (i < n) Console.Write(", ");
18         }
19         Console.WriteLine();
20         Console.ReadKey();
21     }
22 }
23

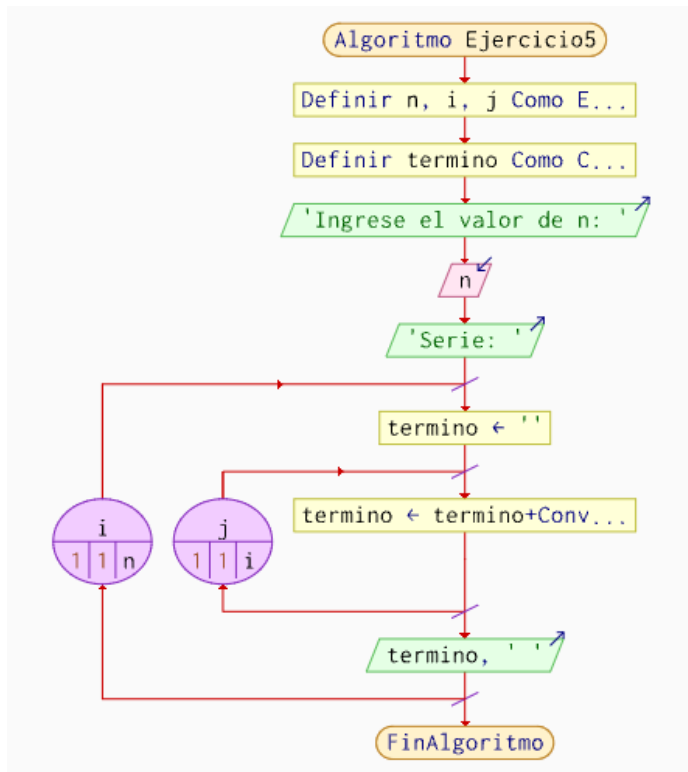
```

Output (C:\Users\USUARIO\Desktop):

```

Ingrese el valor de n: 20
Serie: 1, 22, 333, 4444, 55555, 666666, 7777777,
88888888, 99999999, 101010101010101010, 1111
1111111111111111, 12121212121212121212, 13
13131313131313131313, 14141414141414141414
141414, 15151515151515151515151515, 16161616
161616161616161616161616, 171717171717171717
171717171717, 18181818181818181818181818
18, 19191919191919191919191919191919, 2020
20202020202020202020202020202020

```



6. Imprimir 12345

1234

123

12

1

```

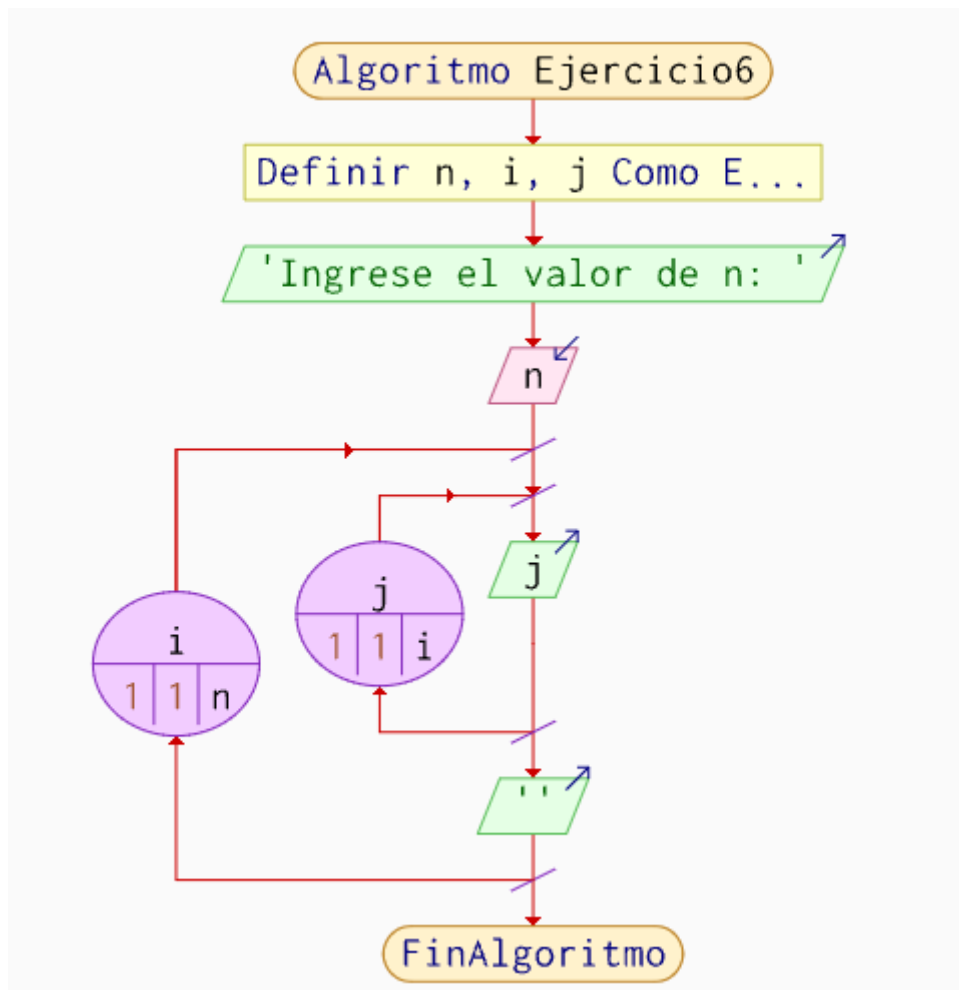
1  using System;
2  0 referencias
3  class Ejercicio6
4  {
5      0 referencias
6      static void Main()
7      {
8          Console.WriteLine("Ingrese el valor de n: ");
9          int n = int.Parse(Console.ReadLine());
10
11          for (int i = 1; i <= n; i++)
12          {
13              for (int j = 1; j <= i; j++)
14                  Console.Write(j);
15              Console.WriteLine();
16          }
17          Console.ReadKey();
18  }

```

```

Ingrese el valor de n: 20
1
12
123
1234
12345
123456
1234567
12345678
123456789
12345678910
1234567891011
123456789101112
12345678910111213
1234567891011121314
123456789101112131415
12345678910111213141516
1234567891011121314151617
123456789101112131415161718
12345678910111213141516171819
1234567891011121314151617181920

```



7.Imprimir \*\*\*\*\*

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```

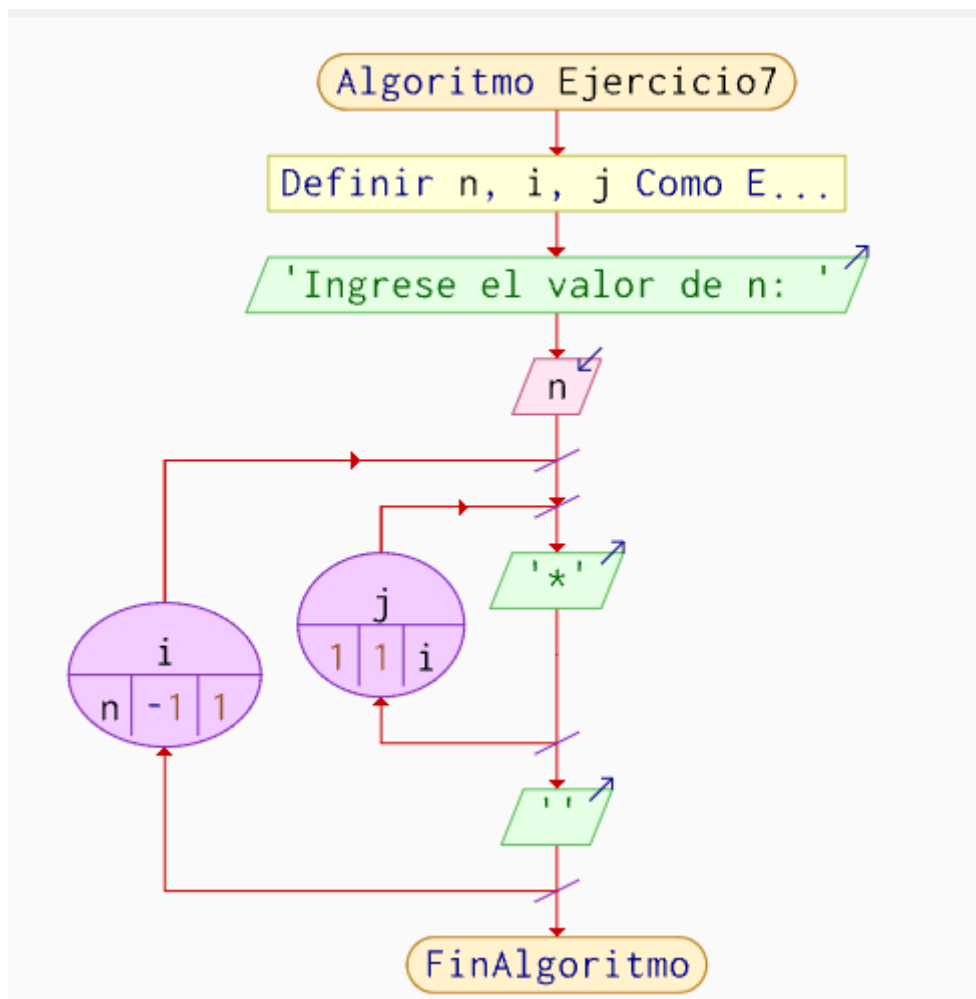
1  using System;
2  0 referencias
3  class Ejercicio7
4  {
5      0 referencias
6      static void Main()
7      {
8          Console.WriteLine("Ingrese el valor de n: ");
9          int n = int.Parse(Console.ReadLine());
10
11          for (int i = n; i >= 1; i--)
12          {
13              for (int j = 1; j <= i; j++)
14                  Console.Write("*");
15
16              Console.WriteLine();
17          }
18          Console.ReadKey();
19      }
  
```

C:\Users\USUARIO\Desktop

Ingrese el valor de n: 5

```

*****
****
***
**
*
  
```



## 8. Imprimir una matriz 2x2

```

1  using System;
2  0 referencias
3  class Ejercicio8
4  {
5      0 referencias
6      static void Main()
7      {
8          int[,] matriz = new int[2, 2];
9          for (int i = 0; i < 2; i++)
10         {
11             for (int j = 0; j < 2; j++)
12             {
13                 Console.WriteLine($"Ingrese el valor [{i},{j}]: ");
14                 matriz[i, j] = int.Parse(Console.ReadLine());
15             }
16             Console.WriteLine("\nMatriz 2x2:");
17             for (int i = 0; i < 2; i++)
18             {
19                 for (int j = 0; j < 2; j++)
20                 {
21                     Console.Write(matriz[i, j] + "\t");
22                 }
23                 Console.WriteLine();
24             }
25             Console.ReadKey();
26         }
27     }
28 }
  
```

Salida de la consola:

```

C:\Users\USUARIO\Desktop>
Ingrese el valor [0,0]: 2
Ingrese el valor [0,1]: 4
Ingrese el valor [1,0]: 5
Ingrese el valor [1,1]: 6

Matriz 2x2:
2      4
5      6
  
```

